

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	A Comprehensive Measure Of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the Human Development Index (HDI) Prediction system. Each feature contributes to providing accurate HDI predictions based on socio-economic indicators and supports users in analyzing a country's development status.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Accepts Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita from the user.	Frontend	Done	High
2	Input Validation	Validates all user inputs before prediction.	Backend	Done	High
3	Data Preprocessing	Cleans and prepares HDI dataset for model training.	Data Processing	Done	High
4	Machine Learning Model	Uses a trained Linear Regression model to predict the HDI score/category.	ML Module	Done	High
5	Prediction Engine	Processes user input and generates real-time HDI prediction.	Backend	Done	High
6	Result Display	Displays the predicted HDI score/category through the web interface.	Frontend	Done	Medium
7	Data Visualization	Generates graphs and charts for HDI data analysis.	Visualization	Done	Medium

8	Flask Web Application	Integrates the trained model with a web interface for user interaction.	Deployment	Pending	High
---	-----------------------	---	------------	---------	------

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Prediction Result Page
2	Backend / Logic	3	Input Validation, Prediction Engine, Result Processing
3	Database / Storage	1	HDI Dataset, Trained Model (Pickle File)
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	1	Input Validation and Exception Handling